

SCHEDULE-VI
[Syllabus of the Examination]

1. Syllabus for General Ability Test Paper-I (Grade-IV):

Sl. No.	Topic		Marks
1	English	English Composition will cover Synonyms, Antonyms, use of common Phrase & Idioms., use of appropriate Prepositions and Articles, Comprehension, ordering of words in a sentence, Ordering of sentences, spotting of errors, use of appropriate and qualifying words. (OMR based MCQ Type - 15 Questions one mark each)	15
2	General Mental Ability & Logical reasoning	Logical Reasoning, Analytical Reasoning Capabilities, Quantitative and Qualitative abilities, General Aptitude (OMR based MCQ Type - 15 Questions one mark each)	15
3	General Knowledge & Current Affairs	(i) General Knowledge with special reference to Tripura and Northeastern States (ii) Elementary knowledge of Indian History and Indian Geography. (iii) Matter of common experience and current events and problems with special reference to Tripura, India and world. (iv) (OMR based MCQ Type & 20 Questions one mark each)	20
Total			50

2. a) Syllabus for Job Oriented Subjects (Grade-IV) Examination:

(i) Technical Paper-II (Full Marks-120)

Engineering Mathematics

Section 1: Probability and Statistics: Random variables. Uniform, normal, exponential, Poisson and binomial distributions. Mean, median, mode and standard deviation. Conditional probability and Bayes theorem.

Computer Science and Information Technology Section 2:

Digital Logic

Boolean algebra, Combinational and sequential circuits, Minimization, Number representations and computer arithmetic (fixed and floating point).

Section 3: Computer Organization and Architecture

Machine instructions and addressing modes, ALU; data path and control unit. Instruction pipelining, pipeline hazards. Memory hierarchy: cache, main memory and secondary storage; I/O interface (interrupt and DMA mode).

Section 4: Analog and Digital Communication

Autocorrelation and power spectral density, properties of white noise, filtering of

random signals through LTI systems, amplitude modulation and demodulation, angle modulation and demodulation, spectra of AM and FM, Super heterodyne receivers, circuits for analog communications, Information theory, entropy, mutual information and channel capacity theorem, Digital communications, PCM, DPCM, digital modulation schemes, amplitude, phase and frequency shift keying (ASK, PSK, FSK), QAM, MAP and ML decoding, matched filter receiver, calculation of bandwidth, SNR and BER for digital modulation; Fundamentals of error correction, Hamming codes; Timing and frequency synchronization, inter- symbol interference and its mitigation; Basics of TDMA, FDMA and CDMA

Section 5: Data Structures and Programming

Arrays, stacks, queues, linked lists, trees, binary search trees, binary heaps, graphs. Programming in C, C++, Java, Recursion.

Section 6: Algorithms

Searching, sorting, hashing. Asymptotic worst-case time and space complexity. Algorithm design techniques: greedy, dynamic programming and divide and conquer. Graph traversals, minimum spanning trees, shortest paths

Section 7: Compiler Design

Lexical analysis, parsing, syntax-directed translation. Runtime environments. Intermediate code generation. Local optimization, Data flow analyses: constant propagation, liveness analysis, common subexpression elimination.

Section 8: Operating System

System calls, processes, threads, inter process communication, concurrency and synchronization.

Deadlock. CPU and I/O scheduling. Memory management and virtual memory. File systems.

Section 9: Databases

ER- model. Relational model: relational algebra, tuple calculus, SQL. Integrity constraints, normal forms. File organization, indexing (e.g., B and B+ trees). Transactions and concurrency control.

Section 10: Information Systems and Software Engineering:

Information gathering, requirement and feasibility analysis, data flow diagrams, process specifications, input/output design, process life cycle, planning and managing the project, design, coding, testing, Implementation, maintenance.

Section 11: Computer Networks

Concept of layering: OSI and TCP/IP Protocol Stacks; Basics of packet, circuit and virtual circuit-switching; Data link layer: framing, error detection, Medium Access Control, Ethernet bridging; Routing protocols: shortest path, flooding, distance vector and link state routing; Fragmentation and IP addressing, IPv4, CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT); Transport layer: flow control and congestion control, UDP, TCP, sockets; Application layer protocols: DNS, SMTP, HTTP, FTP, Email.

Section 12: Web Technologies:

HTML5, CSS3, XML, basic concept of client-server computing, web server, proxy server, web application development, MVC Architecture, web services, frontend technologies.

Section 13: Cyber Security and Emerging Technologies:

Secure programming techniques, OWASP top 10 vulnerabilities, concepts on IoT, Block chain, AI.

Section 14: Cloud Technology:

Cloud Computing, Compute, Network, Storage Management Technologies, Edge Computing.

(OMR based MCQ Type & 120 Questions one mark each)